LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

Quick Revision: Metals and Non-metals

Physical properties

Property	Metals	Non-metals
Lustre	Shiny	Dull (iodine is an exception)
Malleability / ductility	Can be beaten into sheets and drawn into wires	Brittle: break into pieces
Conduction	Good conductors of heat and electricity	Poor conductors (graphite conducts electricity)
Sound	Sonorous (ring when struck)	Not sonorous

Chemical properties

- Metals burn to form **basic oxides**: $2Mg + O_2 \rightarrow 2MgO$. The oxide in water turns **red litmus blue**: $MgO + H_2O \rightarrow Mg(OH)_2$.
- Non-metals form **acidic oxides**: $S + O_2 \rightarrow SO_2$. In water it turns **blue litmus red**: $SO_2 + H_2O \rightarrow H_2SO_3$.
- Sodium reacts vigorously with water, so it is stored in **kerosene**. Phosphorus catches fire in air, so it is stored in **water**.
- Many metals react with dilute acids to give **hydrogen**, which burns with a pop sound: $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$.
- A more reactive metal **displaces** a less reactive metal from its salt solution: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (the blue colour fades).

> Reactivity (most to least, common ones): sodium > magnesium > zinc > iron > copper
> Copper cannot displace zinc or iron, so no reaction happens when copper is put in zinc sulphate.

Uses

- Metals: machinery, cars, utensils, wires (copper, aluminium), jewellery (gold, silver), food wrapping (aluminium foil).
- Non-metals: oxygen for breathing, nitrogen in fertilisers, chlorine to purify water, iodine solution on wounds.

Quick check

1. Why are electric wires made of copper or aluminium?

2. What happens when an iron nail is put in copper sulphate solution? Write the word equation.